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The Mean Value Inequality

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June 15, 2026

0.1 The Mean Value Theorem

Definition 1. Let $(E, ||\cdot||)$ be a normed vector space over $\mathbb{R}$, with subset $A \subseteq E$. A *polygonal line* in A is a sequence of $n+2$ points $u_0, \ldots, u_{n+1}$ in E, such that each line segment connecting neighbouring vertices lies completely in A, i.e.

$$t \cdot u_i + (1-t) \cdot u_{i+1} \in A \quad \forall\, i = 0, \ldots, n \text{ and } t \in [0,\, 1].$$

Lemma 2. *Let U be a subset of a normed vector space. Given a polygonal line in U, joining $x \in U$ and $y \in U$, together with another polygonal line joining y and $z \in U$, then by connecting the two polygonal lines head-to-tail, the result is a polygonal line in U, joining x and z.*

Proof. The construction of the composition of polygonal lines follows common sense: For the polygonal lines a, defined by $(x,\, a_1,\, \ldots,\, a_n,\, y)$, and b, defined by $(y,\, b_1,\, \ldots,\, b_m,\, z)$, we define the composition $a \circ b$ by the sequence $(x,\, a_1,\, \ldots,\, a_n,\, y,\, b_1,\, \ldots,\, b_m,\, z)$.

It remains to show that each segment is still contained in U. This is trivial as the first $n+2$ vertices belong to a and are all contained in U. Similarly, the last $m+2$ vertices belong to b, and are all contained in U as well. □

Theorem 3. *if U is an open connected set of a normed vector space, then any two points of U can be joined by a polygonal line in U.*

Proof. The case where $U = \emptyset$ is trivial. So we assume U to be non-empty. Fix a point $x \in U$, and consider the set

$$V \coloneqq \{u \in U \mid \exists \text{ a polygonal line connecting } x \text{ and } u.\}$$

We want to show that $U = V$.

To show that, we prove that V is both closed and open in U and V is non-empty. The latter is true, as x must be in V.

For closedness, we show that $V = \bar{V}$, the closure of V. Fix a point $a \in \bar{V}$, then $\exists\, \epsilon > 0$, such that $B_\epsilon(a) \cap V \neq \emptyset$. Take a point $b \in B_\epsilon(a) \cap V$. Since $b \in V$, there is a polygonal line joining x and b. Since $b \in B_\epsilon(a)$, there is a line segment joining b and a, which is a polygonal line as well. By composing the two polygonal lines, we create a polygonal line joining x and a. Hence $a \in V$.

For openness, fix a point $a \in V$. We take an open ball centred at a and contained in U, as U is open. It can be shown that this open ball is contained in V as well. It suffices to show that every point b in the open ball can be connected to x using a polygonal line, which can be constructed by composing the polygonal line joining x and a with the line segment joining a and b.

Since V is both open and closed in U, it is either equal to U or is the empty set. Since it is non-empty, we must have $U = V$. □

Definition 4. Let U be an open connected set of a Banach space E. For $a, b \in U$, we define $d_U(a, b)$ to be the infimum of the lengths of the polygonal lines contained in U joining a and b.

Theorem 5. *$d_U(a,b)$ is a distance in the topological space U.*

Proof. Let φ be an arbitrary polygonal line joining a to b in U, defined by the sequence $a_0, \dots, a_{n+1}$, with $a_0 = a$ and $a_{n+1} = b$

Positivity: The length of φ is $\text{Length}(\varphi) := \sum_{i=0}^{n} ||a_{i+1} - a_i||$, which by the definition of a norm function is strictly non-negative. Since $d_U(a,b)$ is the infimum of a strictly non-negative set, it must itself be non-negative. If $d_U(a,b) = 0$, by the previous theorem we have $||b-a|| \le d_U(a,b) = 0$, which implies $a = b$.

Symmetry: For any polygonal line φ from a to b in U, the reverse line ψ defined by $a_{n+1}, \dots, a_0$ is a polygonal line from b to a. $\text{Length}(\psi) = \sum_{i=0}^{n} ||a_i - a_{i+1}|| = \sum_{i=0}^{n} ||a_{i+1} - a_i|| = \text{Length}(\varphi)$, so the set of lengths of polygonal lines from a to b and from b to a are equivalent and therefore $d_U(a,b) = d_U(b,a)$.

Triangle Inequality: Let φ be a polygonal line from a to b and ψ be a polygonal line from b to c. The composition of φ and ψ is a polygonal line θ from a to c, where

$$\begin{cases} \varphi \text{ is defined by } (a, a_1, \dots, a_n, b) \\ \psi \text{ is defined by } (b, b_1, \dots, b_m, c) \\ \theta \text{ is defined by } (a, a_1, \dots, a_n, b, b_1, \dots, b_m, c) \end{cases}$$

By the definition of infimum, $d_U(a,c) = \inf \text{Length}(\theta) \le \text{Length}(\theta) = \text{Length}(\varphi) + \text{Length}(\psi)$. Since φ and ψ are arbitrary, we take the infimum on the right hand side, and hence $d_U(a,c) \le d_U(a,b) + d_U(b,c)$. □

Theorem 6. *Let U be a connected open set of a Banach space E. Let $f : U \to F$ be a differentiable mapping with values in a Banach space F. Assume that*

$$||f'(x)|| \le k \quad \forall x \in U$$

then for any $x_1, x_2 \in U$, $||f(x_1) - f(x_2)|| \le k \cdot d_U(x_1, x_2)$.

Proof. Let $x_1, x_2 \in U$ and l a polygonal line contained in U joining x_1 and x_2. Suppose l is defined by $n+1$ vertices $a_0, \dots, a_{n+1}$, where $a_0 = x_1, a_{n+1} = x_2$ and $a_i \in U$. Then from the mean value inequality on each interval (a_i, a_{i+1}), we have $||f(a_i) - f(a_{i+1})|| \le k \cdot ||a_i - a_{i+1}||$. From the triangle inequality we have

$$\begin{aligned} ||f(x_1) - f(x_2)|| &\le \sum_{i=0}^{n} ||f(a_i) - f(a_{i+1})|| \\ &\le \sum_{i=0}^{n} k \cdot ||a_i - a_{i+1}|| \\ &= k \cdot \text{Length}(l) \end{aligned}$$

as $\text{Length}(l)$ is the sum of the length of each interval. Therefore, taking the infimum we get

$$||f(x_1) - f(x_2)|| \le k \cdot \inf \text{Length}(l) = k \cdot d_U(x_1, x_2)$$

$\square$

Definition 7. Let $(E, ||\cdot||)$ be a normed vector space over $\mathbb{R}$. The *length of an interval* with origin $a \in E$ and end $b \in E$ is $d(a, b) := ||b - a||$.

Definition 8. Given a polygonal line φ defined by $(a_0, \ldots, a_{n+1})$, with some fixed $n \geq 0$ the *length of the polygonal line* φ is the sum of the lengths of the intervals from which it is formed:

$$\text{Length}(\varphi) := \sum_{i=0}^{n} d(a_i, a_{i+1}) = \sum_{i=0}^{n} ||a_{i+1} - a_i||$$

Lemma 9. *Given a sequence* $(a_i)_{i=0}^n$ *with* $a_i \in E$ *for some* $n \geq 1$, $\sum_{i=0}^{n-1} ||a_{i+1} - a_i|| \geq ||a_n - a_0||$

Proof.

- **Base Case ($n = 1$):** For $n = 1$, we need $||a_1 - a_0|| \geq ||a_1 - a_0||$ which follows immediately by reflexivity.
- **Induction Hypothesis:** Assume that the result holds for any such sequence $(b_i)_{i=0}^k$. That is,

$$\sum_{i=0}^{k-1} ||b_{i+1} - b_i|| \geq ||b_k - b_0||$$

- **Induction Step:** We want to show that $\sum_{i=0}^{k} ||a_{i+1} - a_i|| \geq ||a_{k+1} - a_0||$ follows from this assumption. For some arbitrary sequence $(a_i)_{i=0}^{k+1}$,

$$\begin{aligned} ||a_{k+1} - a_0|| &= ||(a_{k+1} - a_k) + (a_k - a_0)|| \\ &\leq ||a_{k+1} - a_k|| + ||a_k - a_0|| \\ &\leq ||a_{k+1} - a_k|| + \sum_{i=0}^{k-1} ||a_{i+1} - a_i|| \\ &= \sum_{i=0}^{k} ||a_{i+1} - a_i|| \end{aligned}$$

Therefore, by the principle of mathematical induction, $\sum_{i=0}^{n-1} ||a_{i+1} - a_i|| \geq ||a_n - a_0||$ for any such sequence with any $n \geq 1$. $\square$

Theorem 10. *The length of a polygonal line* φ *defined by* $a_0, \ldots, a_{n+1}$ *for some fixed* $n \geq 0$ *between origin* $a = a_0$ *and end* $b = a_{n+1}$ *is at least equal to the length of the interval between them:*

$$\sum_{i=0}^{n} ||a_{i+1} - a_i|| \geq ||b - a||$$

Proof. The result follows immediately from the lemma. $\square$